QUIRINO HIGH SCHOOL

TECHPRO • GRADE 11 ACTIVITY SHEET

CSS OFFSETS, Z-INDEX, FLOAT & CLEAR

Positioning offsets • Layering • Text wrapping • Clearing floats

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Learning Objectives

- Use top, right, bottom, and left to adjust positioned elements.
- Explain how z-index controls the stacking order of overlapping elements.
- Use float to place content to the left or right and allow text to wrap.
- Use clear and clearfix to control content after floated elements.

Before You Begin

Use Visual Studio Code and a browser. Create one index.html file, paste the starter code, and refresh the browser after each change.

Starter Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Layout Practice</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 20px; line-height: 1.6; }
    header { background: #6b1024; color: white; padding: 18px; }
    .demo { padding: 12px; margin: 12px 0; border: 2px solid #6b1024; background: #f8e9ed; }
  </style>
</head>
<body>
  <header><h1>TechPro CSS Layout Lab</h1></header>
  <main>
    <h2>Offsets and Layers</h2>
    <div class="demo">Box A</div><div class="demo">Box B</div><div class="demo">Box C</div>
    <h2>Float and Clear</h2>
    <div class="article">
      <div class="photo-placeholder">PHOTO</div>
      <p>Write a short paragraph about a school technology activity. Add enough text to observe how text wraps
around the floated box.</p>
      <p>Continue with more information about the activity, tools used, and what students learned.</p>
    </div>
    <div class="after-float">This paragraph should begin below the floated content.</div>
  </main>
</body>
</html>
```

PART A — Position Offsets: top, right, bottom, left

1. Create a folder named CSS_Offsets_Float_Activity in Visual Studio Code. Save the starter code as index.html.
2. `Add class="offset-box" to Box B. Add CSS: .offset-box { position: relative; left: 25px; top: 10px; }`
3. Save and refresh. Record how Box B moves compared with its original location.
4. Change left to 50px, then try left: -15px. Try bottom: 8px and observe the effect.
5. Offsets work according to positioning mode. A static element is not moved by top, right, bottom, or left.

PART B — z-index: Control Which Box Appears in Front

6. Add a div with class="layer-area", then place three divs inside it with classes layer-one, layer-two, and layer-three. Give each a label.
7. `Add CSS: .layer-area { position: relative; height: 180px; margin: 20px 0; }`
8. `Add CSS: .layer-one { position: absolute; left: 10px; top: 10px; width: 150px; height: 90px; background: #f2c6cf; z-index: 1; padding: 8px; }`
9. `Add CSS: .layer-two { position: absolute; left: 70px; top: 40px; width: 150px; height: 90px; background: #e5b45d; z-index: 2; padding: 8px; }`
10. `Add CSS: .layer-three { position: absolute; left: 130px; top: 70px; width: 150px; height: 90px; background: #b9d9c5; z-index: 3; padding: 8px; }`
11. Observe which box is in front. Change the z-index values so layer-one has the largest value, then refresh.
12. A higher z-index generally appears above a lower z-index within the same stacking context. z-index applies to positioned elements and flex/grid items.

PART C — Float: Wrap Text Around a Box

13. `Add CSS: .photo-placeholder { float: left; width: 110px; height: 90px; margin: 0 14px 8px 0; background: #f2c6cf; border: 2px solid #6b1024; display: flex; align-items: center; justify-content: center; font-weight: bold; }`
14. `Add CSS: .article { border: 1px solid #aaa; padding: 12px; }`
15. Refresh the page. Observe how text flows beside the PHOTO box.
16. Change float: left to float: right and observe how the text wraps.
17. Try float: none. Compare all three results, then restore float: left.

PART D — clear and Clearfix

18. `Add CSS: .after-float { clear: both; margin-top: 16px; padding: 10px; background: #e7edf7; }`
19. Refresh and check that the after-float paragraph starts below floated content.
20. Try clear: left, clear: right, and clear: none. Record what changes.
21. `Add class="clearfix" to the .article container and use: .clearfix::after { content: ""; clear: both; display: table; }`
22. Compare the container before and after clearfix. Describe why clearing is useful.

PART E — Apply the Skills

23. Use offset properties to position a small label on a card.
24. Use z-index to control the order of two overlapping cards.
25. Use float to wrap text around a photo placeholder or school activity image.
26. Use clear or clearfix so the next section does not sit beside or overlap floated content.
27. Test at different browser widths and keep content readable.

MINI-PROJECT — TechPro ICT Showcase

Build a simple school-technology page that combines all four skills.

- Create a page titled TechPro ICT Showcase with a maroon heading and a short school-technology article.
- Add a feature card with a small absolutely positioned FEATURED label; use offset properties to place it.
- Create three overlapping information cards and use z-index to choose which appears on top.
- Add a photo placeholder floated left or right with text wrapping around it.
- Make sure the next section begins below floated content by using clear or clearfix.

CHALLENGE TASK

- Try negative offset values and describe where the element moves.
- Swap z-index values to make a different card appear on top.
- Compare clear: both with the clearfix pseudo-element and explain their purposes.

Check Your Work

☐ My page opens correctly in a browser.

☐ I tested top, right, bottom, and left offsets.

☐ I used z-index to change which card appears in front.

☐ I demonstrated float: left and float: right.

☐ I used clear or clearfix to control the layout after a float.

☐ I tested the page at different widths.

☐ My text remains readable and elements do not unintentionally overlap.

Review Questions

1. What are the four CSS offset properties?

   top, right, bottom, and left

2. Why does position: static not respond to top, right, bottom, or left?

   static is the default positioning mode where elements follow normal document flow and are not moved by offsets

3. What does z-index control?

   controls the stacking orders

4. Which element usually appears in front when two positioned elements in the same stacking context have different z-index values?

   one with a higher value will go infront

5. What does float: left do to an element?

   moves an element to the left side and lets inline content or text flow around it

6. What does float: right do?

   It moves an element to the right side and lets inline content or text flow around it

7. What is the purpose of clear: both?

   prevents the cleared element from sitting beside floats on both the left and right sides

8. What problem can clearfix help solve when a container contains floated children?

   helps a parent container properly contain its floated children so the layout doesn't collapse

9. How does text behave around a floated element?

   Inline content and text wrap or flow around the floated element

10. Give one example of a web layout where float or clear could be useful.

    A layout where text wraps around a photo placeholder, profile image, or a multi-column article block

Quick Review

- Offsets: top, right, bottom, left; use them with an appropriate positioning mode.
- z-index: controls stacking order; larger values appear in front within the same stacking context.
- float: left or right moves an element to a side and lets inline content flow around it; none disables floating.
- clear: left, right, or both prevents the cleared element from sitting beside floats on the specified sides.
- clearfix::after { content: ""; clear: both; display: table; } helps a container contain floated children.

References

- https://www.w3schools.com/css/css_position_offset.asp
- https://www.w3schools.com/css/css_z-index.asp
- https://www.w3schools.com/css/css_float.asp
- https://www.w3schools.com/css/css_float_clear.asp